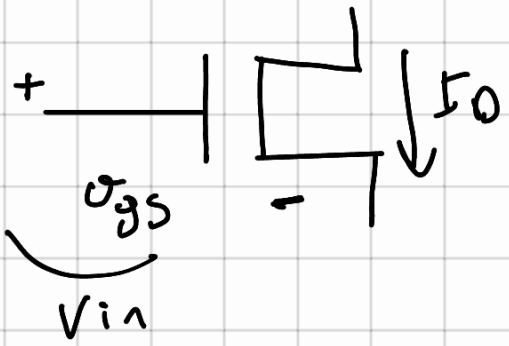
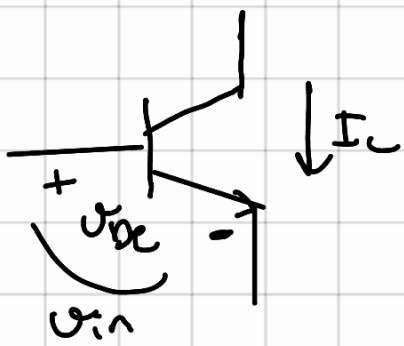
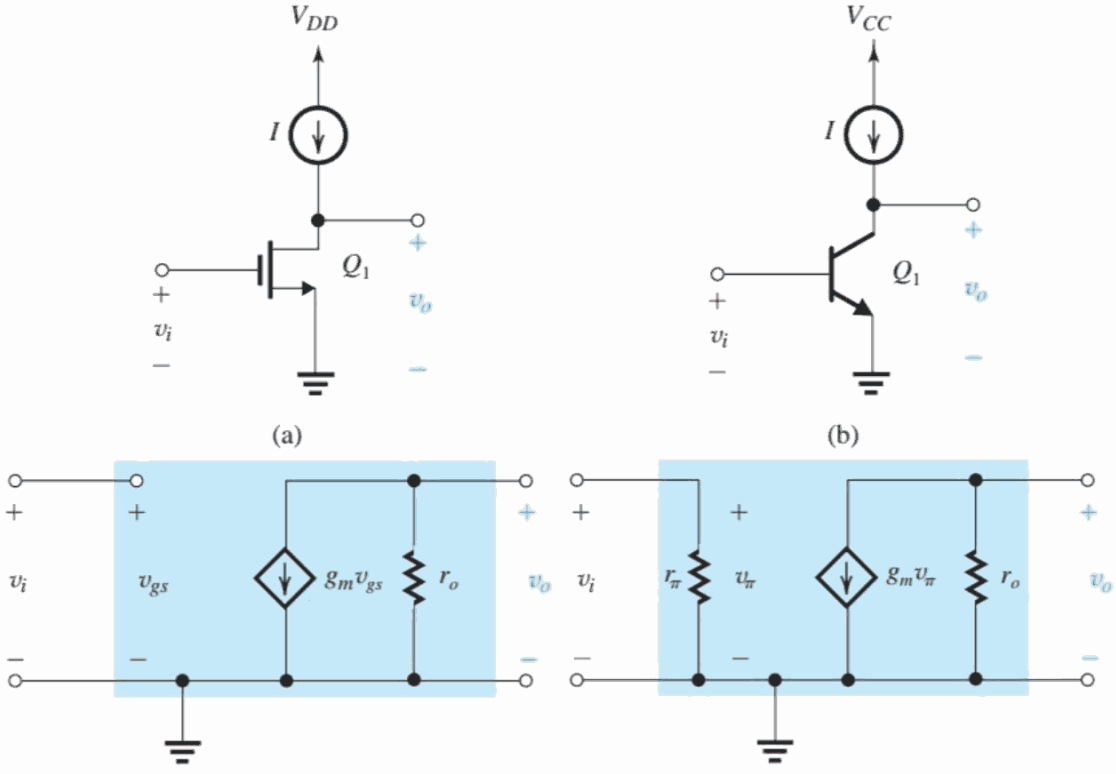








$$R_{in} = \infty \tag{7.1}$$

$$A_{vo} = -g_m r_o \tag{7.2}$$

$$R_o = r_o \tag{7.3}$$

Similarly, from Fig. 7.1(d) we obtain for the active-loaded CE amplifier:

$$R_{in} = r_\pi \tag{7.4}$$

$$A_{vo} = -g_m r_o \tag{7.5}$$

$$R_o = r_o \tag{7.6}$$

1mA, VOV, V_A=0,025

$$A_{v\,(mosfet)} = g_m r_o = \frac{2I_D}{V_{OV}} \cdot \frac{V_A'L}{I_D} = \frac{2V_A'L}{V_{OV}}$$

$$A_{v\,(BJT)} = g_m r_o = \frac{I_C}{V_T} \cdot \frac{V_A}{I_C} = \frac{V_A}{V_T}$$

mosfete göre yüksek

$$I_D = \tfrac{1}{2}\mu_n C_{ox}\frac{W}{L}(V_{OS} - V_{Tn})^2$$

7.2.2 The Intrinsic Gain

For the BJT, we can derive a formula for the intrinsic gain $A_{vo} = g_m r_o$ by using the following formulas for g_m and r_o:

$$g_m = \frac{I_C}{V_T} \tag{7.7}$$

$$r_o = \frac{V_A}{I_C} \tag{7.8}$$

The result is

$$A_0 = g_m r_o = \frac{V_A}{V_T} \tag{7.9}$$

$$g_m = \frac{I_D}{V_{OV}/2} \tag{7.10}$$

$$g_m = \sqrt{2\mu_n C_{ox}(W/L)}\ \sqrt{I_D} \tag{7.11}$$

For the MOSFET r_o we have

$$r_o = \frac{V_A}{I_D} = \frac{V_A' L}{I_D} \tag{7.12}$$

where V_A is the Early voltage and V_A' is the technology-dependent component of the Early voltage. Utilizing each of the g_m expressions together with the expression for r_o, we obtain for A_0,

$$A_0 = \frac{V_A}{V_{OV}/2} \tag{7.13}$$

which can be expressed in the alternate forms

$$A_0 = \frac{2V_A' L}{V_{OV}} \tag{7.14}$$

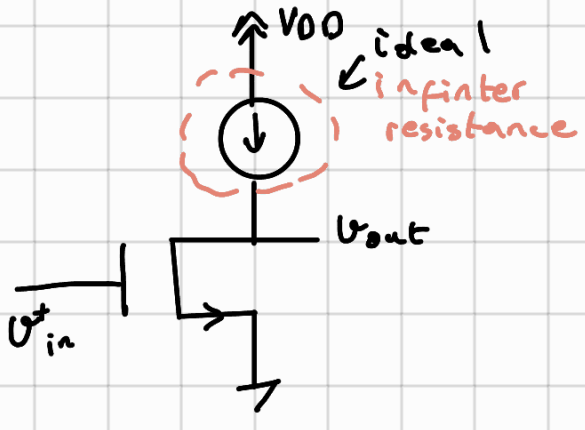
and

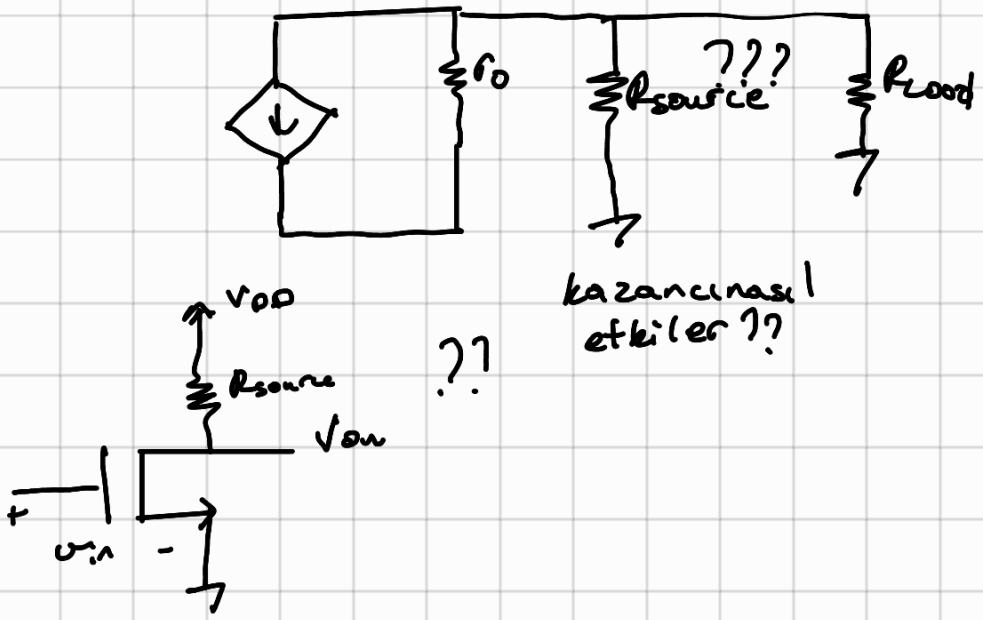
$$A_0 = \frac{V_A'\sqrt{2(\mu_n C_{ox})(WL)}}{\sqrt{I_D}} \tag{7.15}$$

$$I_D = \tfrac{1}{2}\mu_n C_{ox}\frac{W}{L}(V_{OS} - V_{Tn})^2$$

V_{OV}

Gerilim kaynağının içdirenci sıfırdır
Akım kaynağının içdirenci sonsuzdur.





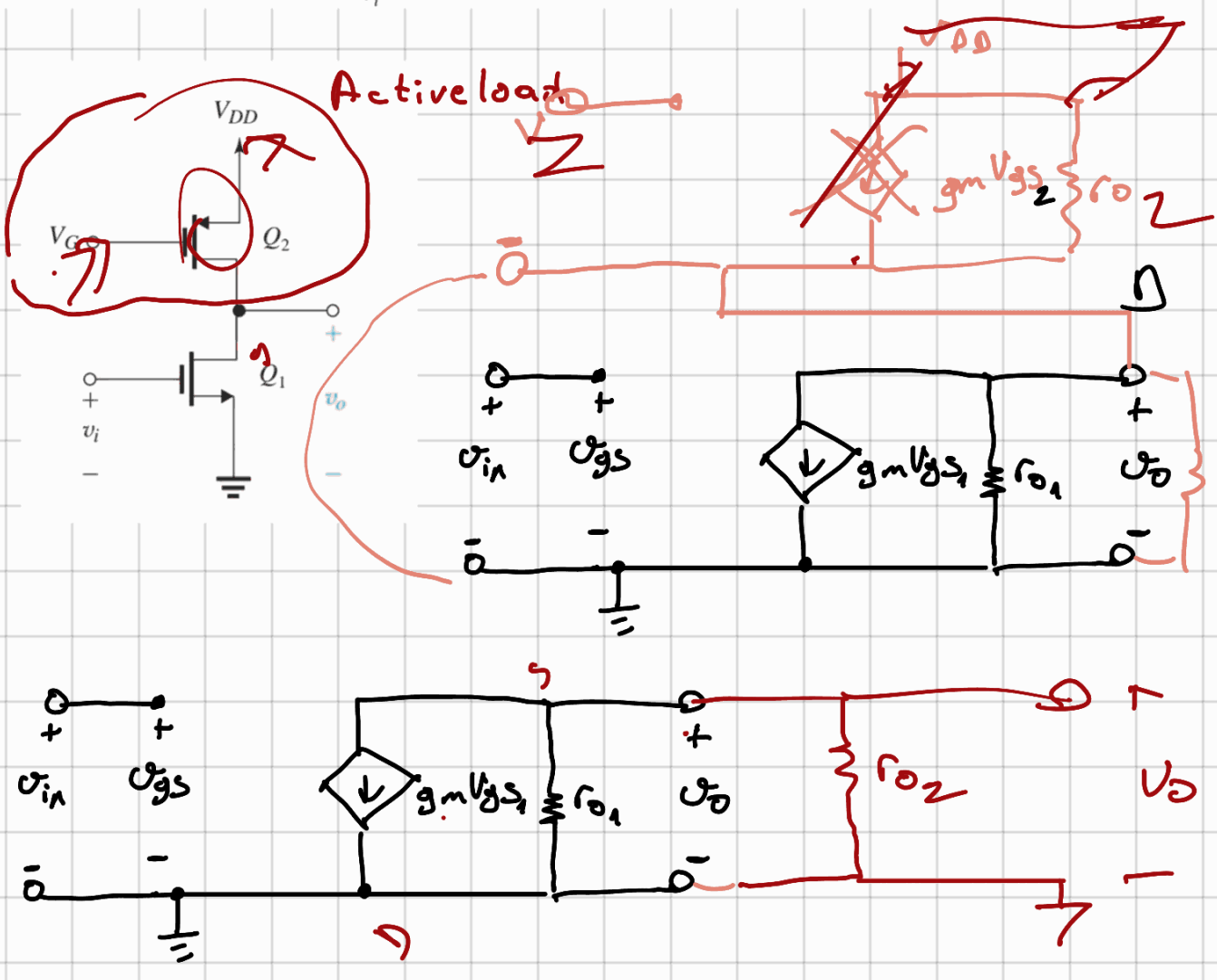
$$I = \frac{1}{2}(\mu_p C_{ox})\left(\frac{W}{L}\right)_2 [V_{DD} - V_G - |V_{tp}|]^2 \tag{7.16}$$

and

$$r_{o2} = \frac{|V_{A2}|}{I} \tag{7.17}$$

Thus the current-source load no longer has an infinite resistance; rather, it has a finite output resistance r_{o2}. This resistance will in effect appear in parallel with r_{o1}, as shown in the amplifier equivalent-circuit model in Fig. 7.3(c), from which we obtain

$$A_v \equiv \frac{v_o}{v_i} = -g_{m1}(r_{o1} \| r_{o2}) \tag{7.18}$$



$$\frac{V_{out}}{V_{in}} = \frac{-g_{m1} V_{gs1} (r_{o1} \| r_{o2})}{V_{gs1}} = -g_m (r_{o1} \| r_{o2})$$

$I_D = r_{o1} = r_{o2} = r_O$ $\quad A_v = -\frac{1}{2} g_m R_S$

$$A_v = \frac{1}{2} \frac{2I_D}{V_{OV}} \frac{V_A' L_1}{I_D}$$

$$\uparrow A_v = \frac{V_A' L_1}{V_{OV1}} \uparrow \downarrow$$

$\downarrow$

$V_{DS} \leq V_{GS} - V_{TH}$ sabit

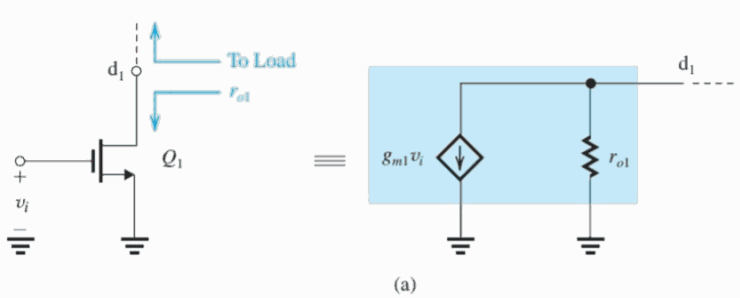
$\downarrow$

Fine tuning

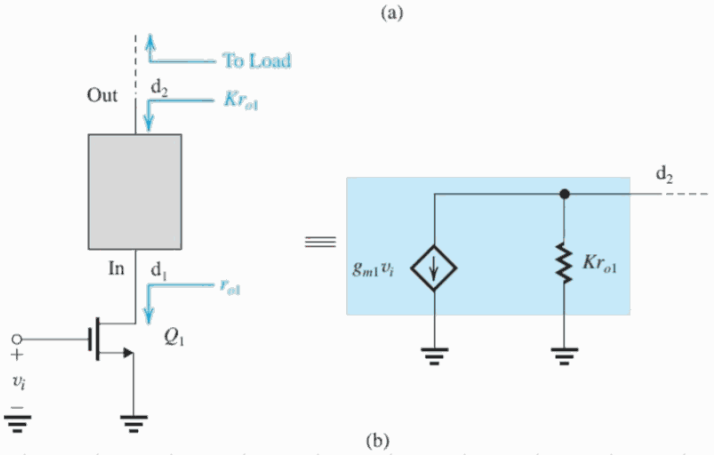
$$I_D = \frac{1}{2} \mu_n C_{ox} \frac{\uparrow W}{\uparrow L} (V_{OV})^2$$

I_D azalma nasıl için

Onun yerine maltistage yapalım



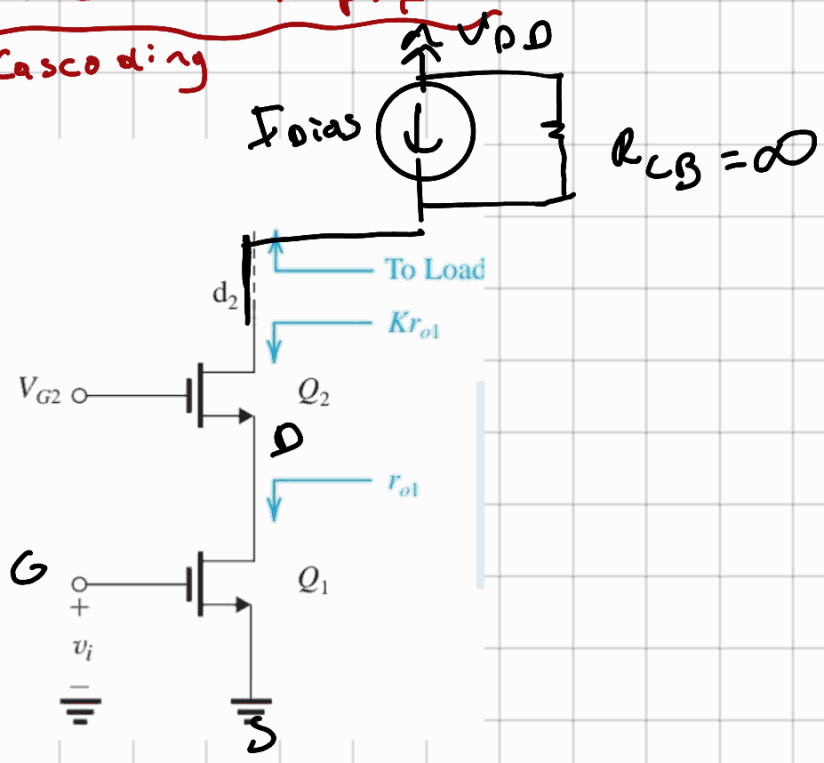
(a)

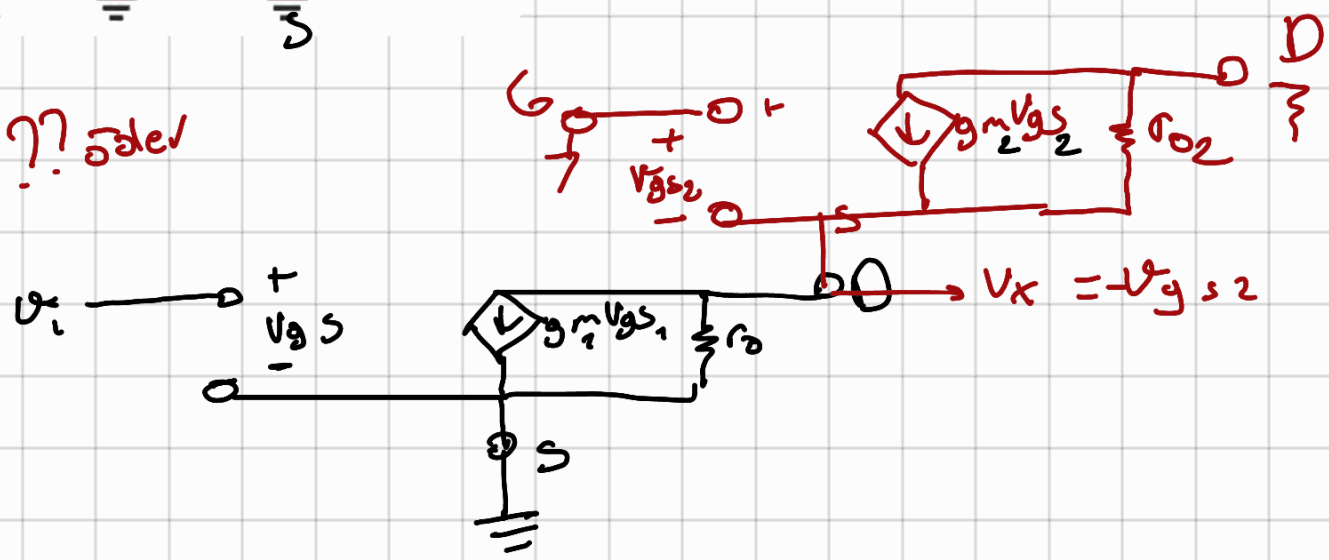


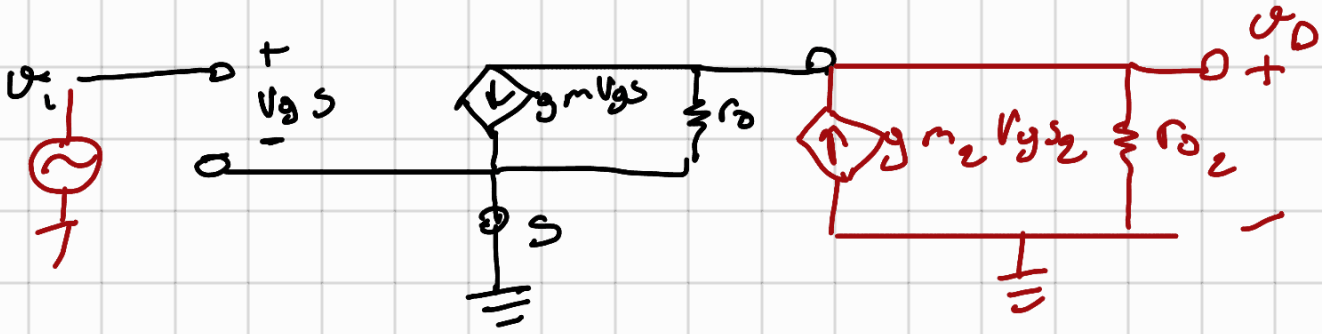
(b)

The Cascode Amplifier

Cascoding

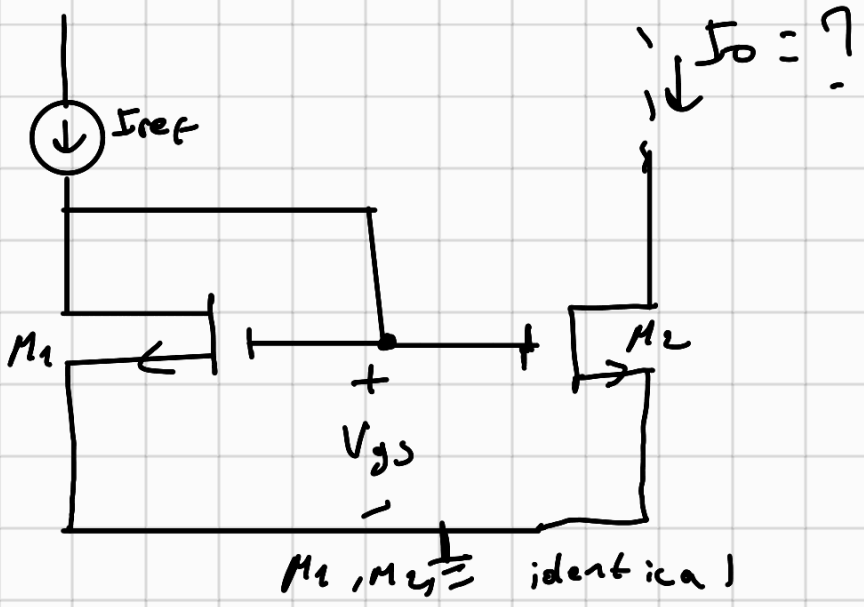






7.4 IC biasing

Current mirror

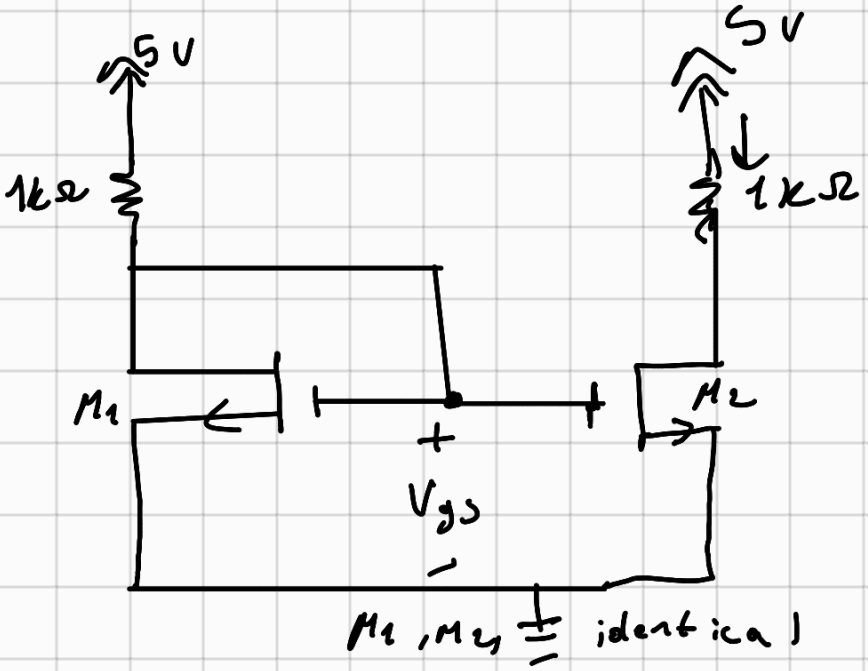


$$I_{D1} = I_{ref} = \frac{1}{2}\mu_n C_{ox}\left(\frac{W}{L}\right)(V_{GS} - V_{TH1})^2(1+\lambda V_{DS1})$$

$$I_{D2} = I_O = \frac{1}{2}\mu_n C_{ox}\left(\frac{W}{L}\right)(V_{GS2} - V_{TH2})^2(1+\lambda V_{DS2})$$

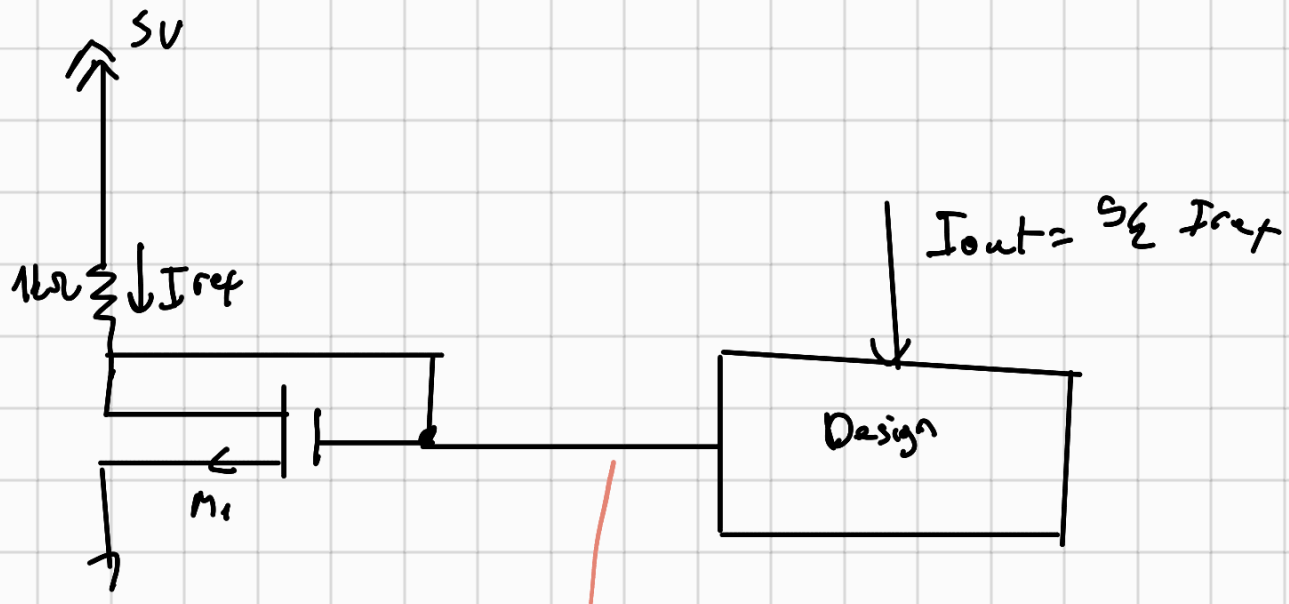
$$\frac{I_{REF}}{I_{O2}} = 1 \qquad I_{REF} = I_{O2}$$

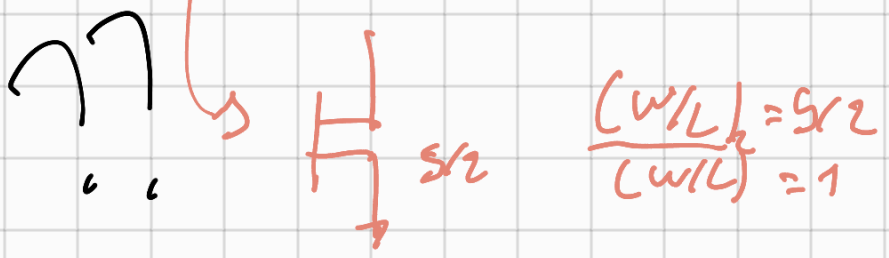
What happen when we put a load ??

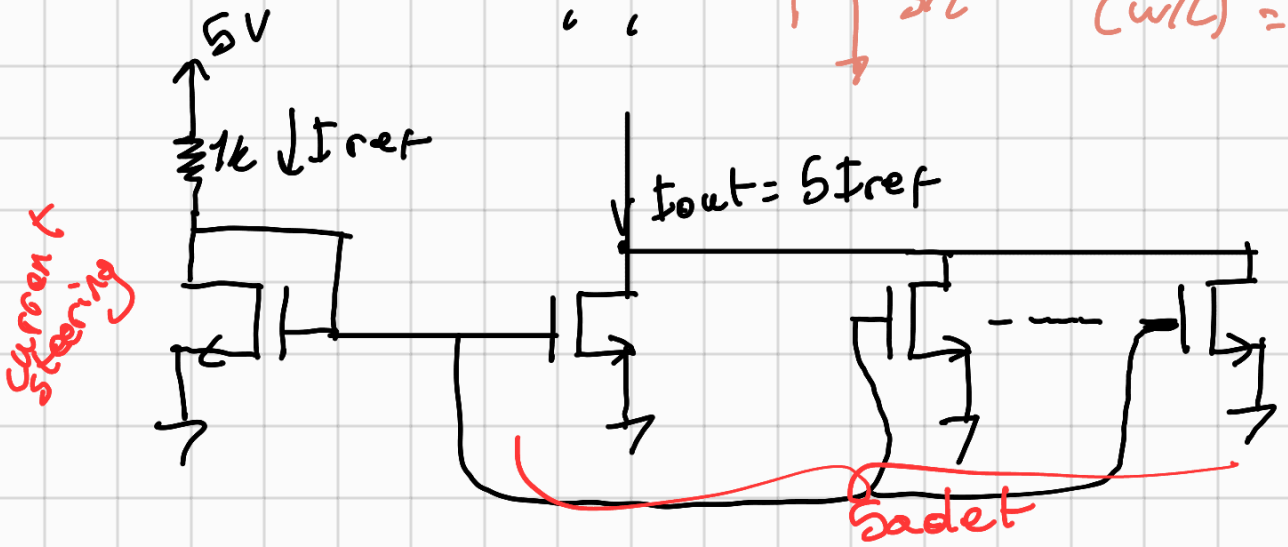


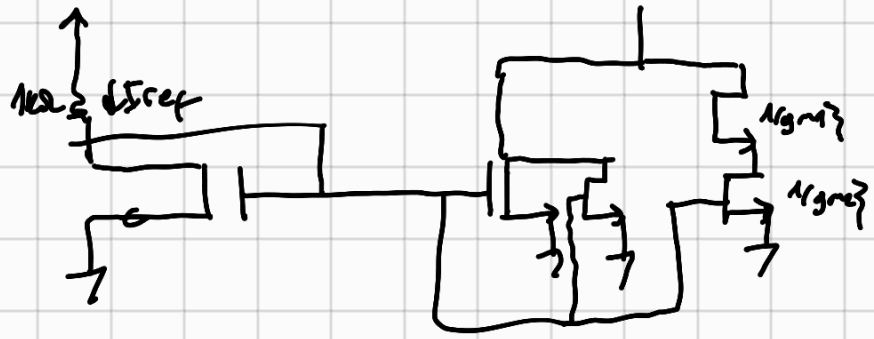
Bu ihtimalde akımlar aynı ve

Yükler aynı ise sadece o gerilimler aynıdır $V_{DS_1} = V_{DS_2}$











7.4.1 The Basic MOSFET Current Source

Figure 7.22 shows the circuit of a simple MOS constant-current source. The heart of the circuit is transistor Q_1, the drain of which is shorted to its gate,[4] thereby forcing it to operate in the saturation mode with

$$I_{D1} = \frac{1}{2}k_n'\left(\frac{W}{L}\right)_1 (V_{GS} - V_{tn})^2 \tag{7.52}$$

where we have neglected channel-length modulation. The drain current of Q_1 is supplied by V_{DD} through resistor R, which in most cases would be outside the IC chip. Since the gate currents are zero,

$$I_{D1} = I_{REF} = \frac{V_{DD} - V_{GS}}{R} \tag{7.53}$$

where the current through R is considered to be the reference current of the current source and is denoted I_{REF}. Equations (7.52) and (7.53) can be used to determine the value required for R.

Now consider transistor Q_2: It has the same V_{GS} as Q_1; thus, if we assume that it is operating in saturation, its drain current, which is the output current I_O of the current source, will be

$$I_O = I_{D2} = \frac{1}{2}k_n'\left(\frac{W}{L}\right)_2 (V_{GS} - V_{tn})^2 \tag{7.54}$$

$$V_O \geq V_{GS} - V_{tn} \qquad (7.56$$

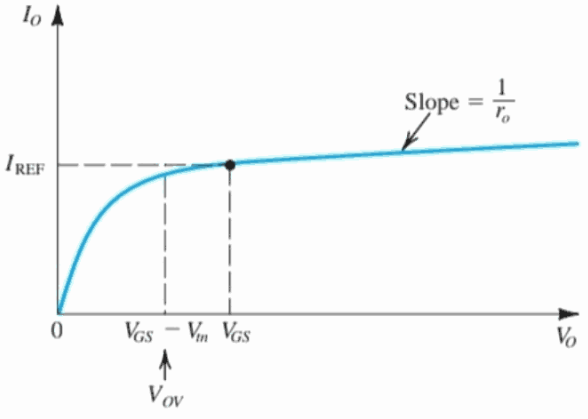
verdrive voltage V_{OV} of Q_1 and Q_2,

$$V_O \geq V_{OV} \qquad (7.57$$

$$R_o \equiv \frac{\Delta V_O}{\Delta I_O} = r_{o2} = \frac{V_{A2}}{I_O} \tag{7.58}$$

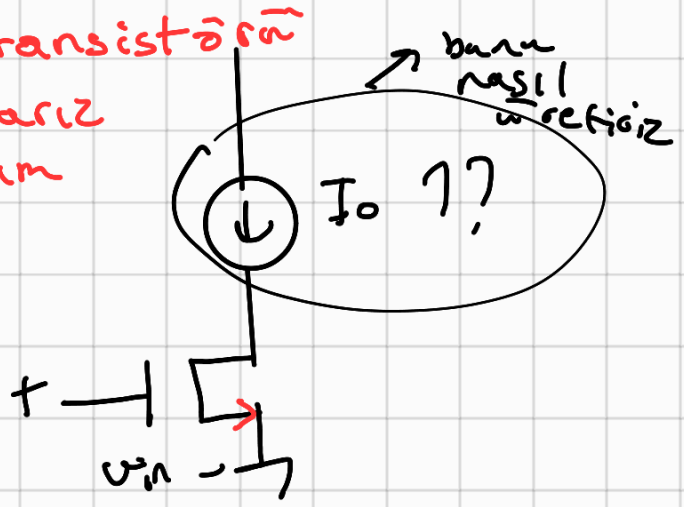
where I_O is given by Eq. (7.54) and V_{A2} is the Early voltage of Q_2. Also, recall that for a given process technology, V_A is proportional to the transistor channel length; thus, to obtain high output-resistance values, current sources are usually designed using transistors with relatively long channels. Finally, note that we can express the current I_O as

$$I_O = \frac{(W/L)_2}{(W/L)_1} I_{REF}\left(1 + \frac{V_O - V_{GS}}{V_{A2}}\right) \tag{7.59}$$

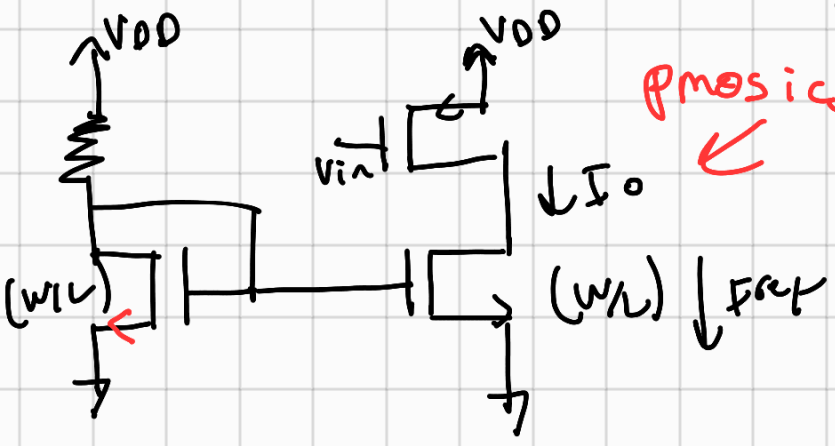


ro sonsuz gidersе
akım sabit olur

bu transistörü nasıl biaslarız ona nasıl akım üretiriz

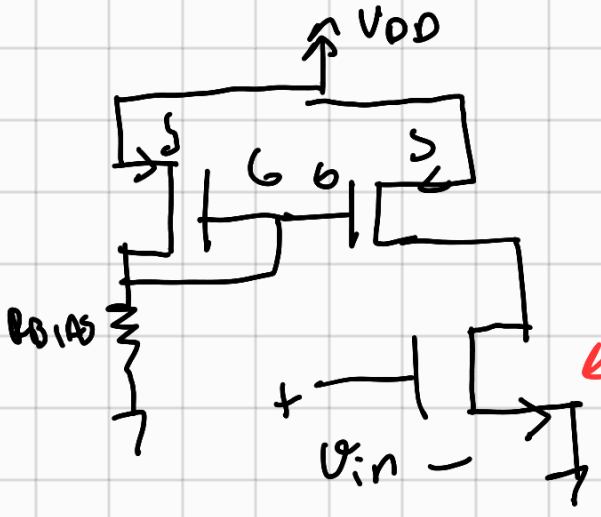


eski



pmos için

geri



nmos için pmoslarla akımı kopya layabiliriz